

DAY 5 TASK REPORT

Extension of LLM vs RAG vs AI agent:

LLM (large language model):

1. Trained huge amount of datasets like docs ,pdfs etc,.
2. In this no excess to real world or live information .
3. We don't use search web ,run code ,like that in this .
4. Input and output are text format only.

How it works:

1. Takes a text input (called a prompt) from the user
2. Processes it through billions of parameters learned during training
3. Predicts and generates the most likely next words (tokens)
4. Produces a text output as a response

RAG (retrieval augmented generation):

1. RAG technique combined llm with knowledge base.
2. RAG first search for relevant information .
3. Can use real, up-to-date data from your own files
4. Does NOT take actions — only retrieves and answers

How it works:

1. User asks a question
2. The system searches a knowledge base (PDFs, databases, websites)
3. Retrieves the most relevant documents or passages
4. Passes those documents along with the question to the LLM
5. LLM generates an answer grounded in the retrieved content

AI agent:

1. It can handle complex ,multiple tasks.
2. Has access to the internet, databases, external APIs
3. It Can learn from previous steps
4. This will work without step-by-step human guidance

How it works:

1. Receives a high-level goal or task from the user
2. Uses the LLM to break the goal into smaller steps (planning)
3. Uses tools — such as web search, code execution, file access, APIs
4. Takes action in the real world based on tool outputs
5. Observes the result and loops back to plan the next step
6. Continues until the task is fully complete